

AVP-Alien Hive, Raid, Contest, and War System

Current system reference for the local AVP-Alien module and shared BLib territory layer

Scope: This document describes the current implemented behavior in the local workspace, not just the design target. AVP-Alien owns alien hive decisions, combat behavior, raids, and lineage rivalry. BLib owns shared player claims, generic territory contest state, claim notifications, and Xaero map/HUD territory display.

1. Executive Snapshot

The current hive system is built around lineages and fixed hive locations. A lineage represents a bloodline, usually descended from a queen and optionally unified under an empress. A hive location is the physical settlement: it owns chunks, stores local reserves, tracks local leadership, runs spawning and biomass growth, and exposes a BLib faction id for map and territory systems.

- Hives are not automatically allies. Different strains fight. Same-strain lineages also fight unless both lineages share the same empress authority.
- Territory ownership is stored in AVP-Alien hive location records and mirrored into BLib territory claims under location faction ids.
- Player claims live in BLib. AVP-Alien asks BLib whether a chunk is player-claimed and can start contests against those claims through BLib.
- Raids are currently player-retaliation convoys: an empress-led lineage tracks player kills, drains reserves, and sends raid waves toward the target player.
- Territory war is border-driven: alive, uninhibited hives scan neighboring chunks, find player claims or hostile hive claims, and start BLib contests.
- Contested chunks are represented client-side as chunks with more than one claimant. BLib makes them blink red in Xaero and warns player claim owners through server messages/actionbar reminders.

2. Main Concepts

Concept	What it means in the current system
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Variant faction	A broader faction bucket for an alien variant/strain. Variant factions are used for organization, naming, aesthetics, and map style backfill.
Lineage faction	One queen/bloodline. It owns one or more hive locations, stores variant, dimension, founder, empress id, convoy list, and player kill attribution.
Hive location	A fixed settlement belonging to exactly one lineage. It owns claimed chunks, reserves, biomass, jelly, local leader state, boss bar, slab height, and local membership maps.
Location faction	The BLib faction id for a specific hive location. This is the id registered as the actual chunk claimant in BLib territory.
Player claim	A BLib-owned chunk claim tied to a player UUID and a player-claim faction id. Shared by Alien, Human, and Predator modules.
Territory contest	A BLib record for attacker vs defender over one chunk. Progress moves toward attacker or defender based on power scoring.
Reserve	An abstract stored alien count or identity entry inside a hive location. Reserves can later materialize as spawned entities or be drained into convoys.
Convoy	An in-flight group of reserves. Current convoy types include reinforcements, migrations, and raids.
Empress authority	The only current rule that prevents hive-vs-hive rivalry between different lineages. Same empress means unified/allied.
Inhibited location	A contained/severed breeder-style location. It can still defend via aggro, but autonomy such as claims, biomass, spawning, and contests is suppressed.

3. Hive System

Data Ownership

The hive system deliberately separates bloodline identity from physical territory. LineageFactionData is the persistent home for the bloodline and stores its nested HiveLocation records. HiveLocationRegistry rebuilds in-memory indexes from those records when factions load.

- LineageFactionData stores variant, dimension, founder id, empress id, location records, active convoys, kill attribution, and lineage death/removal state.
- HiveLocation stores the fixed center, claimed chunks, chunk claim ticks, decorated chunks, biomass, royal/scourge jelly, reserves, known/loaded members by type, leader state, and removal state.
- HiveLocationRegistry indexes locations by id, lineage, chunk, and spatial bucket. These indexes are not the save format; they are rebuilt from factions.
- HiveLocationClaims is the atomic claim/release path. It updates the location record, the in-memory registry, and BLib territory together.

Server Tick Flow

1. Alien.tickHiveRegistry runs once per server tick from the Overworld tick hook so the hive registry drives all dimensions from one place.
2. HiveLocationRegistry ticks every live location: age advances, leader and boss bar logic update, loaded territory work runs, and inhibited locations stop before autonomous growth/spawning behavior.
3. Loaded locations run fast work when at least one claimed chunk is loaded: trespasser aggro, loaded biomass income, catch-up growth, and abstract spread attempts.
4. The slow-path scheduler samples unloaded locations with a bounded per-tick budget, applies unloaded biomass catch-up, passive claims, and abstract spread attempts.
5. Convoys tick every server tick, including raid target tracking, warning messages, arrival/materialization, return-home handling, boss bars, and interception checks.
6. Periodic tasks handle spawning, reinforcements, migrations, raid dispatch, empress emergence, territory war scans, population pressure decay, and lifecycle cleanup.

Growth, Claims, and Spawning

Hive growth is biomass and population gated. A location earns biomass from unloaded abstract income and loaded activity, then spends biomass to claim nearby chunks. Claim expansion respects per-location and per-lineage caps, population requirements, reproductive establishment for queen-founded hives, and angry boss-bar lockout.

- New claims are picked from neighboring territory candidates and must remain connected back to the location center.
- Disconnected claims are pruned so a hive cannot keep isolated islands after losing intermediate chunks.
- A queen-founded hive waits for reproductive establishment before spending biomass on expansion or loaded reserve spawning.

- Loaded spawns draw from local reserves, require nearby players, stay inside the hive slab band, respect minimum/maximum player distance, and favor core chunks for queens, harbingers, praetorians, and crushers.
- Local reserve spawning is capped by configured loaded xenomorph target and per-location spawn attempt limits.

Territory Defense

Every second, loaded hive xenomorphs can be retargeted onto players standing inside that hive location's claimed chunks. This defense still runs for inhibited locations; inhibition blocks autonomous growth and war behavior, not the queen/entity's own combat.

4. Raid System

Current raid meaning: Raids are not the same thing as territory contests. A raid is a moving convoy sent after a player who has built up enough kill credit against an empress-led lineage. Territory contests are chunk ownership fights handled through BLib.

How a Raid Starts

7. When a player kills members of a lineage, LineageFactionData records timestamps for that player inside the raid aggro window.
8. RaidDispatch scans lineages every 5 seconds through the registry dispatch cadence.
9. Only living lineages with an empress id can auto-dispatch raids.
10. If an online player in the same dimension has at least the configured kill threshold, the lineage tries to dispatch a raid.
11. The source is the largest eligible live hive location with at least the configured minimum chunk count, enough reserves for the active raid wave profile, and no active source cooldown.
12. The raid drains its full composition from the source reserves, creates a Convoy.Raid, clears consumed kill attribution, and starts traveling toward the player's last known position.

Default tuning	Current value
Raid kill threshold	5 lineage-member kills
Raid aggro window	10 minutes
Minimum source size	8 claimed chunks
Per-source cooldown	20 minutes

Raid speed multiplier	1.5x convoy speed
Warning lead time	60 seconds before arrival
Dispatch scan cadence	Every 5 seconds through HiveLocationRegistry

Raid Travel and Arrival

- Outbound raids keep updating the player target position while the player is online, alive, and in the raid dimension.
- About one minute before arrival, the player receives a queen scream and the message: A distant screech answers your violence...
- The target receives Marked for Death while the raid is active or inbound.
- On arrival, the raid materializes one wave at the current raid position. The convoy remains alive so missing or despawned raiders can return to the convoy composition.
- If the player dies, the raid returns home when possible or disbands if no live return location exists.
- If the player becomes unavailable because of creative/spectator state, the raid can wait and resume hunting when the target becomes valid again.
- If all raid composition and materialized members are gone, the raid is defeated and grants the defeat-raid advancement to the target player.

Raid Profiles

Raid wave composition is data-driven through raid wave profiles. The active registry picks a profile for the lineage variant and uses weighted pools plus guaranteed groups to drain reserves into waves. Normal, Nether, and Aberrant profiles are generated in the Fabric data provider, with a fallback profile available if a variant-specific profile is missing.

5. Contest and War System

BLib Contest Engine

BLib owns the generic territory contest engine. A contest can start only when the attacker and defender factions are hostile and the defender currently claims the chunk. Starting a contest adds the attacker as another claimant on that chunk, which makes the chunk contested for sync/UI purposes.

- BLib stores active contests by dimension, chunk, attacker, defender, progress, reason, and last update tick.

- Every 100 ticks, BLib compares attacker power and defender power. The power difference moves progress, capped to 10 points per tick pass.
- At +100 progress, the attacker wins. At -100 progress, the defender wins.
- When a contest resolves, BLib removes the loser claim. If the loser is a player claim faction, BLib removes the player claim itself.
- Listeners receive start, progress, resolve, and cancel events. AVP-Alien listens so hive chunks can be released or claimed on resolution.

Alien War Rule

Rule: Alien hives are territorial rivals unless unified under the same empress. Different strain is always hostile. Same strain does not imply alliance; same-strain lineages are enemies unless both lineages share the same empress id.

Situation	Current behavior
Normal hive beside Nether hive	Hostile. They can fight, start a border contest, and take chunks.
Two normal hives, different lineages, no shared empress	Hostile. Same strain is still rival if the bloodline is different and no empress unifies them.
Two hives under the same empress	Unified/allied. They should not start hive-vs-hive contests against each other.
Hive beside a player claim	Hostile relationship is created against the player-claim faction and the hive can contest the chunk.
Inhibited hive location	Cannot project war. It is skipped for contest-start scans.
Queenless/weak hive	Can lose territory if another hostile hive applies stronger contest power.

How Alien Border Contests Start

13. AlienTerritoryWarSystem scans living hive locations at the configured contest window cadence.
14. A location can project war if it is alive, not inhibited, has claimed chunks, and belongs to a living lineage with either empress authority, at least one queen, or enough reliable local reserves.
15. For each owned chunk, the system checks the four cardinal neighboring chunks.
16. If the neighbor is a player claim, AVP-Alien marks the hive location and lineage hostile to that player-claim faction and asks BLib to start a contest.

17. If the neighbor is another live hive location, AVP-Alien compares lineages. Rival lineages become hostile; shared-empress lineages become allied.
18. If hostile and no identical contest already exists, BLib starts the contest with reason `avp_alien:territory_war`.

Alien Contest Power

AVP-Alien registers a BLib contest power provider. The current power score is intentionally simple and local: it rewards actual loaded alien presence in the contested chunk, plus a hive's ability to project pressure from reserves, territory size, and queen presence.

Power source	Effect
Loaded xenomorphs in the contested chunk	Each matching-lineage xenomorph contributes strong immediate power.
Local reserves	A hive with more reliable reserve population projects more pressure if it owns or borders the contested chunk.
Claimed territory size	Larger territory adds some pressure, representing colony scale.
Queen presence	A location with a tracked queen adds extra pressure.
Player claim owner nearby	Player claims have base power and gain more if the owner is standing in or near the contested chunk.
No projection path	A hive that neither owns nor borders the target mostly contributes only actual loaded presence.

What Happens When a Hive Contest Resolves

- If a hive location loses a chunk it owned, AVP-Alien releases that chunk through `HiveLocationClaims`, preserving center-chunk protection and disconnected-claim pruning.
- If a hive location wins a chunk it did not own, AVP-Alien claims it through `HiveLocationClaims` so the location record, registry index, and BLib territory stay synchronized.
- If a player claim loses, BLib removes the player claim. AVP-Alien then claims the chunk for the winning hive if the winner is a hive location.
- Resolve events log winner, loser, chunk, dimension, and the `territory_war` reason for useful debugging.

6. Player-Facing Warnings and Map Behavior

- Xaero map overlays show normal territory with synced faction color or claim-map overlay texture. Contested chunks blink red by alternating opacity and forcing Xaero cache refresh through the region hash.
- Xaero chunk tooltip shows Claim: <username> for player claims, CONTESTED - Claim: <username> for contested player claims, and faction/hive names for non-player territory.
- BLib client HUD/actionbar shows Claim: <label> while standing in claimed territory and CONTESTED - Claim: <label> when standing in contested territory.
- Player claim owners receive server-side warnings even without Xaero installed. Contest start sends a chat/system warning plus actionbar, ongoing contests remind every 30 seconds, and resolve/cancel sends held/lost/ended messages.
- Hive territory also has AVP-Alien render/debug support through the hive render cache and inspector paths, while BLib remains the shared territory UI layer.

7. Current Boundaries and Gotchas

- Raids and territory contests are separate systems today. Raids chase players based on kill attribution; border contests fight over chunks. A raid can interact with enemy territory through normal gameplay, but raid dispatch is not currently the generic chunk-objective attack planner.
- Same strain is not a friendship check. The only implemented same-lineage peace rule for different hives is shared empress authority.
- Player claims are shared BLib state now. AVP-Alien legacy player claims are migrated/read forward into BLib on faction load.
- BLib decides generic contest validity and resolution. AVP-Alien decides alien hostility, pressure scoring, border scan targets, and hive consequences after a contest resolves.
- Inhibited locations intentionally do not expand, spawn reserves, accrue autonomous growth, or project territory war. They can still defend/trespass aggro and the queen entity itself can still fight or lay eggs through entity logic.
- Hive location center chunks are protected from normal release so a location does not accidentally lose its root chunk during ordinary contest cleanup.

8. Useful Debug and Verification Points

Need to verify	Where to look
Hive registry state	/avp_alien debug hive commands and HiveLocationRegistry logs.

Nearest hive/location status	/avp_alien nearest and debug hive inspection/status commands.
Raid force/testing path	/avp_alien debug hive force_raid or raid_from_nearest_hive.
Claim growth	HiveLocationClaims, CatchUpEngine, ChunkPicker, and claim_radius debug helper.
Contest start logs	Alien territory war started log line with attacker, defender, chunk, dimension, and reason.
Contest resolution logs	Alien territory war resolved log line with winner, loser, chunk, and dimension.
Player claim warnings	BLibPlayerClaimContestNotifier server messages/actionbar reminders.
Xaero contested blink	BLibChunkHighlighter contested red blink and ClientTerritoryCache faction list size > 1.
Alien-vs-alien hostility	AlienPredicates.areAliensEnemies plus AlienTerritoryWarSystem.areAlienLineagesEnemies.
Compiled cross-loader behavior	Run common, NeoForge, and Fabric compile tasks after API changes.

9. Source Files Consulted

These are the main files this reference was based on in the current local workspace:

- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\Alien.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\common\gameplay\hive\location\HiveLocationRegistry.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\common\gameplay\hive\location\HiveLocation.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\common\gameplay\hive\faction\LineageFactionData.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\common\gameplay\hive\growth\HiveLocationClaims.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\common\gameplay\hive\growth\CatchUpEngine.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\common\gameplay\hive\convoy\RaidDispatch.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-\common\src\main\java\com\alien\common\gameplay\hive\tick\LineageConvoyTickTask.java

- C:\Users\shayc\IdeaProjects\AVP-Alien-
common\src\main\java\com\alien\common\gameplay\hive\war\AlienTerritoryWarSystem.java
- C:\Users\shayc\IdeaProjects\AVP-Alien-
common\src\main\java\com\alien\common\util\AlienPredicates.java
- C:\Users\shayc\IdeaProjects\BLib-
common\src\main\java\com\blib\api\common\territory\v1\TerritoryContestManager.java
- C:\Users\shayc\IdeaProjects\BLib-
common\src\main\java\com\blib\internal\common\territory\BLibTerritoryContestManager.java
- C:\Users\shayc\IdeaProjects\BLib-
common\src\main\java\com\blib\internal\common\territory\BLibPlayerClaimManager.java
- C:\Users\shayc\IdeaProjects\BLib-
common\src\main\java\com\blib\internal\common\territory\BLibPlayerClaimContestNotifier.java
- C:\Users\shayc\IdeaProjects\BLib-
common\src\main\java\com\blib\internal\client\territory\BLibClaimHud.java
- C:\Users\shayc\IdeaProjects\BLib-
common\src\main\java\com\blib\internal\client\territory\compat\xaero\BLibChunkHighlighter.java